

University of Michigan

Programming for Everybody (Getting Started with Python)

by University of Michigan

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Course Info

About this Course

This course aims to teach everyone the basics of programming computers using Python. We cover the basics of how one constructs a program from a series of simple instructions in Python. The course has no pre-requisites and avoids all but the simplest mathematics. Anyone with moderate computer experience should be able to master the materials in this course. This course will cover Chapters 1-5 of the textbook "Python for Everybody". Once a student completes this course, they will be ready to take more advanced programming courses. This course covers Python 3.

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Taught by:

 Charles Russell Severance

Clinical Professor, School of Information

Basic Info

Course 1 of 5 in the Python for Everybody Specialization

Level

Beginner

Commitment

2-4 hours/week

Language

English

Subtitles: Arabic, French, Bengali, Uzbek, Ukrainian, Chinese (Simplified), Greek, Italian, Portuguese (Brazil), Vietnamese, Dutch, Korean, German, Pashto, Urdu, Russian, Thai, Indonesian, Swedish, Turkish, Azerbaijani, Spanish, Dari, Hindi, Japanese, Kazakh, Hungarian, Polish

How To Pass

Pass all graded assignments to complete the course.

User Ratings

★ 4.8

Syllabus

Module 1

Chapter One - Why We Program

This module contains course-wide materials as well as the first part of Chapter One where we explore what it means to write programs. Throughout the course, you may want to come back and look at these materials. In the third module, we will return to Chapter One and submit the first assignment.

5 videos, 3 readings ^

1. Video: Welcome to Class - Dr. Chuck

2. Video: Welcome from the Creator of Python - Guido van Rossum

3. Reading: Course Overview and Resources

4. Reading: Pre-Course Survey

5. Video: Why Program

6. Video: Hardware Overview

7. Graded Assignment: Programming Concepts Check-in

8. Video: Python as a Language

9. Reading: Industry Voices and Office Hours

10. Role Play: Explore: The Programmer's Role and Impact

View Less ^

Module 2

Installing Python

In this module you will set things up so you can write Python programs.

1 video, 5 readings ^

1. Reading: Important: Using Python in this Class

2. Reading: Choosing a Text Editor

3. Video: Demonstration: Using the Python Playground

4. App Item: Python Code Playground

5. Reading: Installing and Using Python - Windows/macOS

6. Reading: Taking Screen Shots for Assignments - Windows/macOS

7. Graded Assignment: Installing Python

8. Reading: Industry Voices: Eben Upton - Raspberry Pi

Module 3

Chapter One: Why We Program (continued)

In the first chapter, we try to cover the "big picture" of programming so you get a "table of contents" of the rest of the book. Don't worry if not everything makes perfect sense the first time you hear it. This chapter is quite broad and you would benefit from reading the chapter in the book in addition to watching the lectures to help it all sink in. You might want to come back and re-watch these lectures after you have finished a few more chapters.

4 videos, 2 readings ^

1. Video: Elements of Python

2. Video: Writing Paragraphs of Code

3. Video: Demonstration: Doing the "Hello World" Assignment

4. Reading: Submitting Programming Assignments

5. Reading: Industry Voices: Daphne Koller - Building Coursera

6. Video: Fake-to-Fake Office Hours: Milan, Italy

Graded: Quiz: Chapter 1

Graded: Assignment: Write Hello World

Module 4

Chapter Two: Variables and Expressions

In this chapter, we cover how a program uses the computer's memory to store, retrieve and calculate information.

7 videos, 3 readings ^

1. Video: Constants, Reserved Words & Variables

2. Video: Variable Names and Assignment

3. Video: Numerical Expressions

4. Video: Variable Types

5. Graded Assignment: Expressions Check-in

6. Video: Writing Comments in Python

7. Video: Your First I/P/O Program

8. Reading: Where Is the Worked Exercise for Assignment: Welcome Message?

9. Reading: Worked Exercise: Pay Calculator

10. Reading: Industry Voices: Proop Sanjkar - Building Piazza

11. Video: Office Hours: Mountain View, CA

Graded: Quiz: Chapter 2

Graded: Assignment: Welcome Message

Graded: Assignment: Pay Calculator

Module 5

Chapter Three: Conditional Code

In this section we move from sequential code that simply runs one line of code after another to conditional code where some steps are skipped. It is a very simple concept - but it is how computer software makes "choices".

4 videos, 3 readings ^

1. Video: Conditional Statements, Part 1

2. Video: Conditional Statements, Part 2

3. Video: Conditional Statements, Part 3

4. Reading: Worked Exercise: Exercise 3.2

5. Reading: Industry Voices: Massimo Banzì - Arduino

6. Video: Office Hours: Seoul, South Korea

Graded: Quiz: Chapter 3

Graded: Assignment: Overtime Pay Calculator

Graded: Assignment: Write Conditional Statements

Module 6

Chapter Four: Functions

This is a relatively short chapter. We will learn about what functions are and how we can use them. The programs in the first chapters of the book are not large enough to require us to develop functions, but as the book moves into more and more complex programs, functions will be an essential way for us to make sense of our code.

4 videos, 1 reading ^

1. Video: Using Functions

2. Video: Building Functions, Part 1

3. Video: Building Functions, Part 2

4. Reading: Industry Voices: Guido van Rossum - the Early Years of Python

5. Video: Office Hours: Manila, Philippines

Graded: Quiz: Chapter 4

Graded: Assignment: Build Functions

Module 7

Chapter Five: Loops and Iteration

Loops and iteration complete our four basic programming patterns. Loops are the way we tell Python to do something over and over. Loops are the way we build programs that stay with a problem until the problem is solved.

7 videos, 5 readings ^

1. Video: Loops and Iteration

2. Video: Definite Loops

3. Video: Finding the Largest Value

4. Graded Assignment: Loops and Iterations Check-in

5. Video: Loop Idioms

6. Video: Finding the Smallest Value

7. Reading: Worked Exercise: Exercise 5.1

8. Reading: Industry Voices: Guido van Rossum - the Modern Era of Python

9. Video: Office Hours: Paris, France

10. Video: What's Next - Dr. Chuck

11. Reading: Please Rate this Course on Class-Central

12. Reading: Post-Course Survey

13. Reading: Keep Learning With Michigan Online!

Graded: Quiz: Chapter 5

Graded: Assignment: Find the Largest and Smallest Numbers

How It Works

General

What do start dates and end dates mean?

Once you enroll, you'll have access to all videos, readings, quizzes, and programming assignments (if applicable). If you choose to explore the content without purchasing, you may not be able to access certain assignments. If you don't finish all graded assignments before the end of the session, you can reset your deadlines. Your progress will be saved and you'll be able to pick up where you left off.

What are due dates? Is there a penalty for submitting my work after a due date?

Within a course, there are suggested due dates to help you manage your schedule and keep work from piling up. Quizzes and programming assignments can be submitted late without consequence. However, it is possible that you won't receive a grade if you submit your peer-graded assignment too late because classmates usually review assignment within three days of the assignment deadline.

Can I re-attempt an assignment?

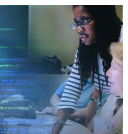
Yes. If you want to improve your grade, you can always try again. If you're re-attempting a peer-graded assignment, re-submit your work as soon as you can to make sure there's enough time for your classmates to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review learning material during this delay.

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Course 1 of Specialization

Learn to Program and Analyze Data with Python

Develop programs to gather, clean, analyze, and visualize data.

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Python for Everybody

View the course in catalog

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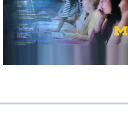
Python Data Structures

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Using Python to Access Web Data

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Using Databases with Python

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Capstone: Retrieving, Processing, and Visualizing Data with Python